ES32079 Financial Derivatives
Problem Set 1

Question 1

Give an example of a **hedger**, a **speculator**, and an **arbitrageur** in derivatives markets.

Why might a hedger sometimes be accused of "speculating"'? What controls should firms have in place?

Question 2

Which of the following instruments are OTC vs exchange-traded?

- Futures on crude oil
- A customized forward contract on jet fuel
- An S&P 500 index option
- A 5-year interest rate swap

What are the advantages and disadvantages of OTC vs exchange? Why do most FX and interest rate derivatives trade OTC instead of on exchanges?

Question 3

A farmer expects to harvest 50,000 bushels of wheat in 3 months. Wheat futures (contract size 5,000 bushels) for delivery in 3 months trade at \$5.20 per bushel.

1. What hedge should the farmer take today?
2. How many contracts are required?
3. If the spot price turns out to be \$4.80 at harvest and the futures settle at \$4.85, what is the effective price the farmer receives?

Why is the hedge not perfect?

Question 4

Define the **basis** and explain why it matters in a hedge.

Spot price of corn today = \$3.00, futures price (3 months) = \$3.10. In 3 months, spot = \$2.70, futures = \$2.75.

- What was the basis at hedge initiation and at hedge close?
- How did the basis risk affect the hedge outcome?

Why does the basis tend to converge to zero at maturity?

Question 5

An airline wants to hedge jet fuel costs, but there are no jet fuel futures. It considers using crude oil futures.

What are the risks in this strategy? What factors determine whether cross-hedging is effective?

Question 6

Historical regression of ΔS on ΔF gives slope $h^* = 0.8$. If the airline's fuel exposure is 1,000,000 gallons and each futures contract = 42,000 gallons, how many contracts should it use?

Why might h^* be less than 1? Can over-hedging ever make sense?